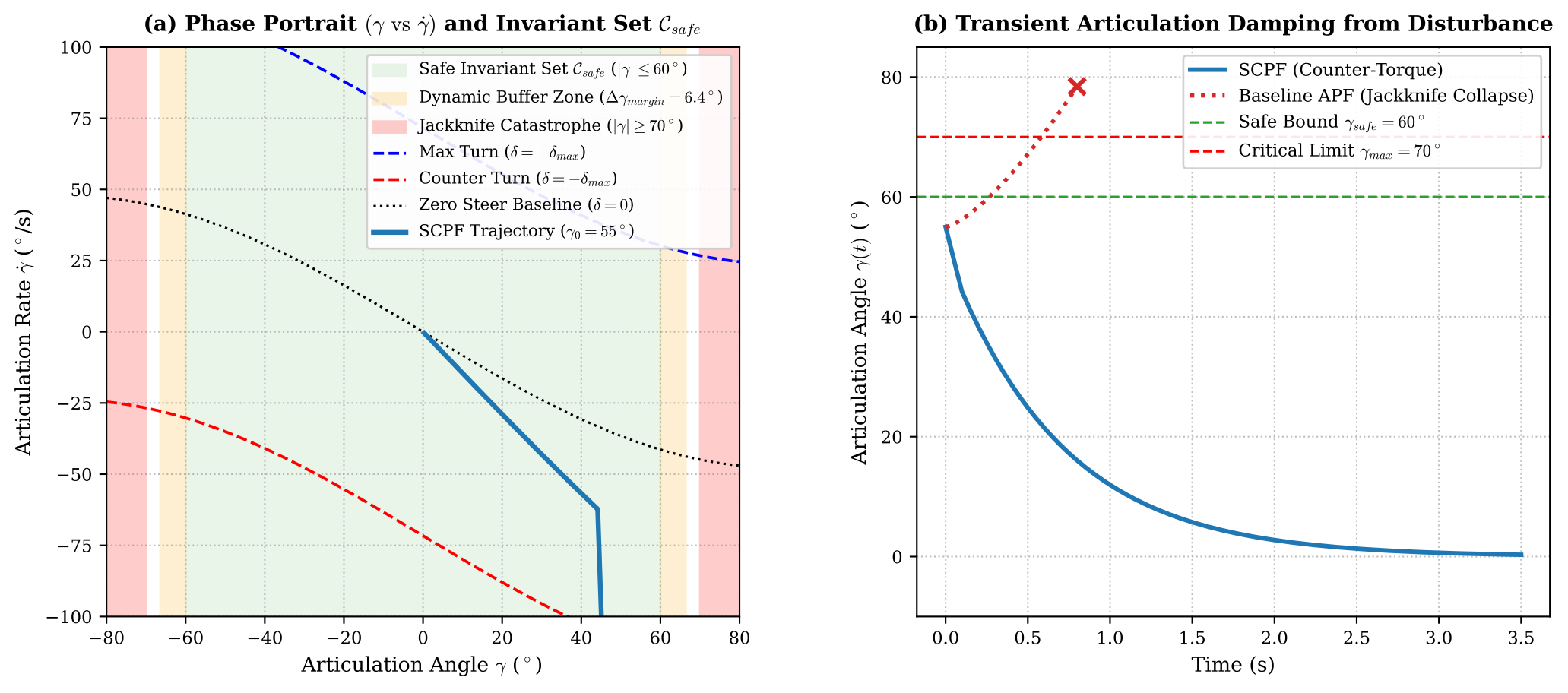

(a) Phase Portrait (γ vs $\dot{\gamma}$) and Invariant Set $\mathcal{C}_{safe}$
Safe Invariant Set $\mathcal{C}_{safe}$ ($|\gamma| \leq 60^\circ$)
Dynamic Buffer Zone ($\Delta\gamma_{margin} = 6.4^\circ$)
Jackknife Catastrophe ($|\gamma| \geq 70^\circ$)
Max Turn ($\delta = +\delta_{max}$)
Counter Turn ($\delta = -\delta_{max}$)
Zero Steer Baseline ($\delta = 0$)
SCPF Trajectory ($\gamma_0 = 55^\circ$)
Articulation Rate $\dot{\gamma}$ ($^\circ$/s)
Articulation Angle γ ($^\circ$)
100
75
50
25
0
−25
−50
−75
−100
−80
−60
−40
−20
0
20
40
60
80
(b) Transient Articulation Damping from Disturbance
SCPF (Counter-Torque)
Baseline APF (Jackknife Collapse)
Safe Bound $\gamma_{safe} = 60^\circ$
Critical Limit $\gamma_{max} = 70^\circ$
Articulation Angle $\gamma(t)$ ($^\circ$)
Time (s)
80
60
40
20
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5